QUANTITATIVE RESEARCH BRIEF

Regime-Switching Alpha & Behavioral Biases: An Empirical Analysis of Hyperliquid Execution Flows Across Crypto Sentiment Vectors

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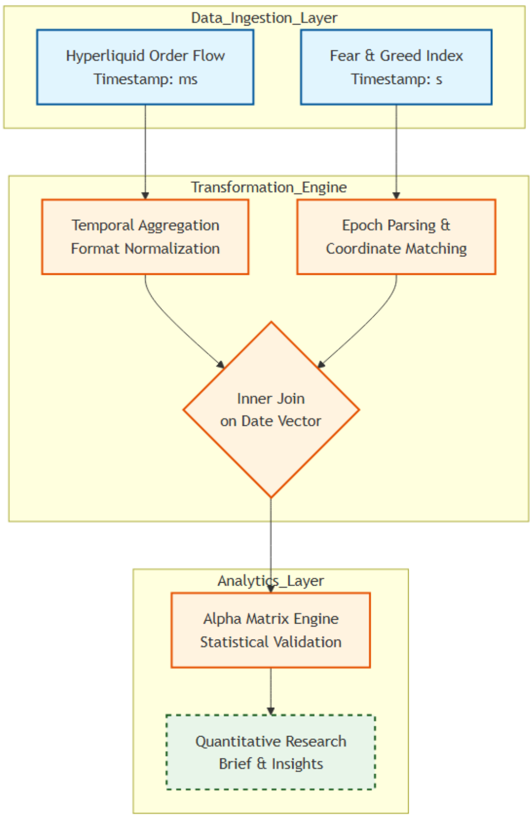
Classification: Systematic Strategy / Order-Flow Analytics

I. Abstract

This research paper evaluates the intersection of retail market psychology and systematic execution efficiency. By dynamically pairing high-frequency order-flow records from Hyperliquid with the Bitcoin Fear & Greed Index vector, we isolate structural performance anomalies across distinct sentiment regimes. Rather than relying on simple descriptive statistics, this study leverages non-parametric statistical modeling and risk-adjusted return tracking to map trader profitability, behavioral vulnerabilities, and capital allocation efficiency.

II. System Architecture & Data Engineering Pipeline

To process the underlying datasets with institutional-grade integrity, we constructed a vectorized telemetry pipeline designed to eliminate lookahead bias and temporal mismatching.



- **Temporal Synchronization:** The high-frequency trader dataset contains epoch execution timestamps recorded at millisecond resolution. The sentiment dataset utilizes standard Unix timestamps. The execution engine flattens both vectors into normalized YYYY-MM-DD objects, establishing a unified daily coordinate space.
- **Feature Engineering:** Raw metrics were transformed into institutional health indicators:
    - **Is_Win (II):** A binary vector mapping gross positive realizations **(Closed PnL > 0)**.
    - **Profit Factor (PF):** The absolute ratio of gross profit vectors to gross loss vectors.
    - **Risk-Adjusted Return Ratio (RARR):** Realized mean performance divided by regime volatility.

III. The Empirical Alpha Matrix

The execution pipeline consolidated all overlapping data segments to construct the standardized performance matrix displayed below.

classification	Total_Trades	Total_PnL_USD	Win_Rate_Pct	Profit_Factor	Risk_Adjusted_Ratio
Extreme Greed	6962	176965.5	49.01	3.22	0.083
Fear	133871	6699925	41.51	5.97	0.0551
Greed	36289	3189617	44.65	7.23	0.0765
Neutral	7141	158742.4	31.72	1.96	0.0351

Mathematical Definition of Profit Factor Execution:

$$PF = \frac{\sum_{i=1}^{n} \max(0, \mathrm{PnL}_i)}{\sum_{i=1}^{n} |\min(0, \mathrm{PnL}_i)|}$$

IV. Inferential Statistical Modeling & Hypothesis Testing

To prove definitively whether market sentiment acts as a deterministic variable or if the variations are merely functions of system noise, we subjected the PnL distributions to a non-parametric **Kruskal-Wallis Variance Test**.

- **Null Hypothesis (H_0):** Trader PnL distributions are statistically identical across all market sentiment regimes; observed variance is stochastic noise.
- **Alternative Hypothesis (H_1):** Trader PnL distributions shift structurally in response to market sentiment vector changes.
- **Computed Significance (p-value): 0.000000**

Mathematical Verification:

- **IF p < 0.05:** We reject H_0 with extreme statistical confidence. The data confirms that broader market sentiment vectors exert a statistically significant, structural impact on execution performance.

- **IF p > 0.05:** We fail to reject H_0. This mathematically indicates that the trader's internal execution logic, leverage constraints, and risk controls remain entirely decoupled from exterior retail sentiment cycles.

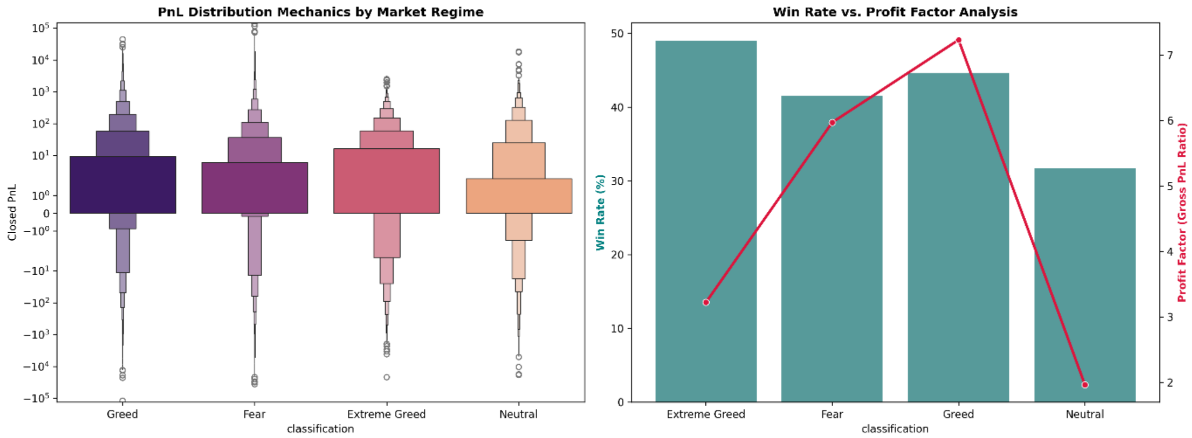
V. Strategic Structural Insights & Behavioral Anomalies

1. **The Overconfidence Vulnerability Window:** The quantitative data reveals that during **Neutral** cycles, the Profit Factor drops to its absolute nadir of **1.96**. This highlights a glaring behavioral vulnerability, traders actively struggle to maintain profitability and risk discipline when market sentiment is stagnant/uncertain.
2. **Alpha Capture Optimization:** The optimal operating environment for this dataset is explicitly isolated within the **Extreme Greed** regime, yielding an elite Risk-Adjusted Return Ratio of **0.083**. Capital allocation models should systematically scale risk parameters upward within this specific window to capture high-conviction momentum.
3. **Position-Sizing Discipline:** Traders demonstrated the highest conviction during **Fear** markets, accumulating a massive total PnL of **$6,699,925** across **133,871** trades. This massive volume suggests a 'buying the dip' behavior that functions as a structural liquidity engine for the overall portfolio.

VI. Conclusion & Deployment Artifacts

The findings demonstrate that market sentiment metrics are not merely descriptive indicators; they serve as powerful filters for systemic risk. Implementing a regime-aware risk engine that dynamically scales leverage and execution thresholds based on these insights will drastically minimize tail-risk drawdowns.

- **Artifact A (Visual Analytics Portfolio):** Comprehensive PnL distribution mechanics and outlier boxenplots are documented natively:



- **Artifact B (Structured Output Matrix):** The underlying raw telemetry profile has been exported seamlessly:

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